MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

DRIVER, ANALOG, KEY CODE MAD

This specification is approved for use within the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic analog driver module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787	-	Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1331	-	Parameter to be Controlled for the Specification of Microcircuits.
MIL-STD-1389	-	Design Requirements for Standard Electronic Modules.
MIL-STD-1665	-	Test Equipment for the Standard Electronic Modules Program.

(Copies of the specification and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Space and Naval Warfare Systems Command, ATTN: SPAWAR 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.25 failure per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be MAD.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 20 contact pins positioned A1 through A20. Contact pins in positions A21 through A40 are optional.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

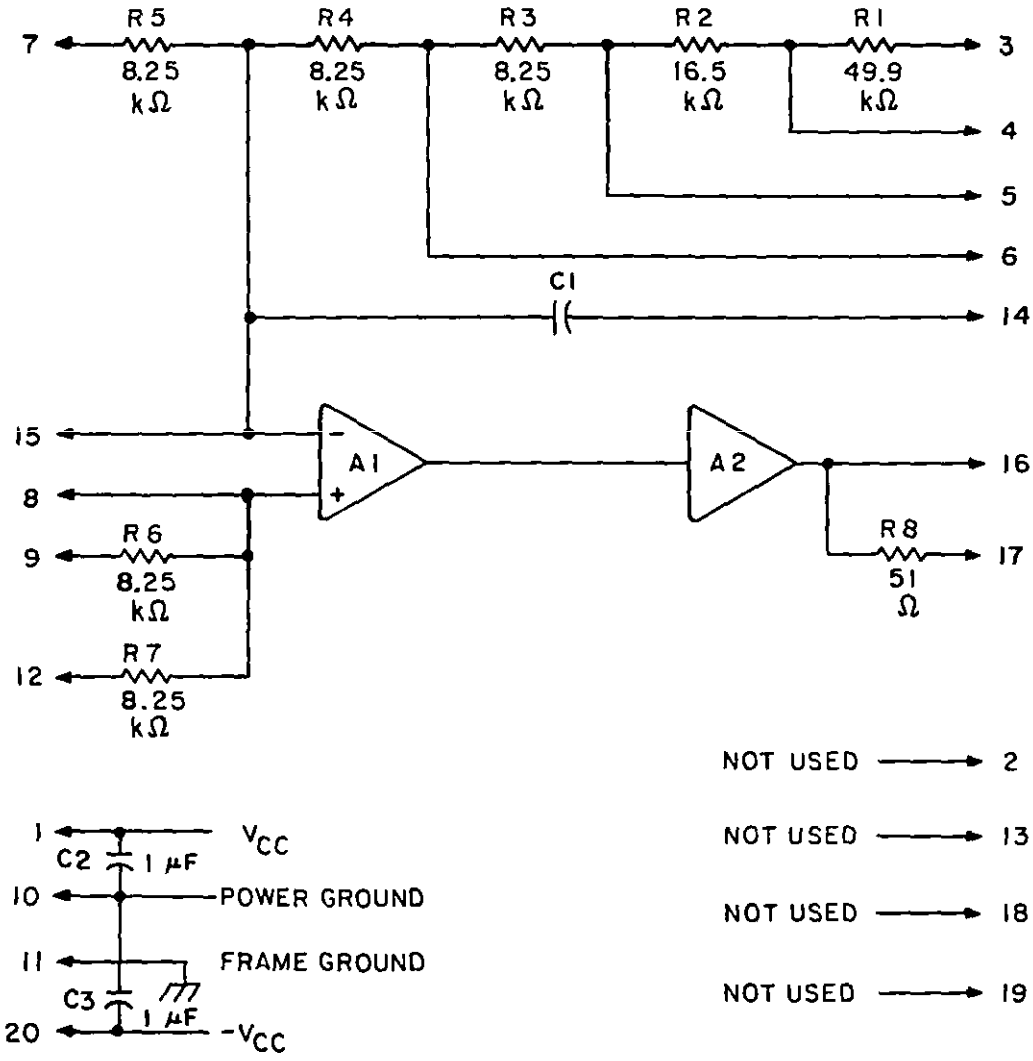
3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/202-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

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NOTES:
1. Resistors R1 through R7 are ±1.4%, 1/8 W.
2. Resistor R8 is ±15%, 1/4 W.
3. Capacitor C1 is ±20%.
4. Capacitors C2 and C3 are ±25%.
5. Specified tolerances reflect the operating temperature 0°C to 60°C end-of-life limits from table I.

* FIGURE 1. MAD module function diagram, driver.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1	Output short circuit current	II	S	30	300	30	300	30	300	30	300	mA
A2	Output short circuit current	II	S	30	300	30	300	30	300	30	300	mA
B1-B5	Isolation	III	X	10	∞	1	∞	10	∞	1	∞	MΩ
C1, C2	Power supply current	IV	S	1	39	1	39	1	42	1	42	mA
D	Input bias current check	V	X	0.5	140.0	0.5	140.0	0.5	140.0	0.5	140.0	nA
E1, E2	Input offset voltage check	VI	X	-3.5	3.5	-3.5	3.5	-4.5	4.5	-4.5	4.5	mV
F	Input offset current check	VI	X	-50	50	-50	50	-50	50	-50	50	nA
G1, G5	Inverting gain check	VII	X	9.75	10.35	9.69	10.41	9.69	10.41	9.67	10.43	V/V
G2, G6	Inverting gain check	VII	X	3.88	4.12	3.856	4.144	3.856	4.144	3.848	4.152	V/V
G3, G7	Inverting gain check	VII	X	1.940	2.060	1.928	2.072	1.928	2.072	1.924	2.076	V/V
G4, G8	Inverting gain check	VII	X	0.970	1.030	0.964	1.036	0.964	1.036	0.962	1.038	V/V
H	Inverting frequency response check	VIII	X	-4.0	-2.4	-4.2	-2.2	-4.2	-2.2	-4.4	-2.0	dB
J	Noninverting gain check	IX	X	0.98	1.02	0.98	1.02	0.98	1.02	0.98	1.02	V/V
K	Noninverting phase margin check	IX	S	45	180	45	180	45	180	45	180	Degree
L	Noninverting bandwidth	IX	S	0.43	3.00	0.43	3.00	0.43	3.00	0.43	3.00	MHz

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
M1	Distortion	X	X	0	1	0	1	0	1	0	1	%
M2	Distortion	X	X	0	1	0	1	0	1	0	1	%
N1	Oscillation check	XI	X	0	5	0	5	0	5	0	5	mVp-p
N2	Oscillation check	XI	X	0	600	0	600	0	600	0	600	mVp-p
N3	Oscillation check	XI	Q	0	5	0	5	0	5	0	5	mVp-p
N4	Oscillation check	XI	Q	0	600	0	600	0	600	0	600	mVp-p
P1, P2	Power supply capacitance	XII	X	0.75	1.25	0.75	1.25	0.75	1.25	0.75	1.25	µF
R1	Resistance	XIII	X	49.40	50.40	49.25	50.55	49.25	50.55	49.20	50.60	kΩ
R2	Resistance	XIII	X	16.33	16.67	16.29	16.71	16.29	16.71	16.27	16.73	kΩ
R3-R7	Resistance	XIII	X	8.167	8.333	8.143	8.357	8.143	8.357	8.134	8.366	kΩ
R8	Resistance	XIII	X	48.45	53.55	45.90	56.10	45.90	56.10	43.40	58.60	Ω
S1	Large signal voltage gain check	XIV	X	20	400	20	400	20	400	20	400	V/mV
S2	Large signal voltage gain check	XIV	X	50	400	50	400	32	400	32	400	V/mV
T	Input impedance check	XV	S	1	∞	1	∞	0.5	∞	0.5	∞	MΩ
- - -												

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
U	Broadband noise	XVI	S	0	2	0	2	0	2	0	2	mV rms
V	Supply voltage rejection ratio check	XVII	S	-∞	-40	-∞	-40	-∞	-40	-∞	-40	dB
W	Common mode rejection ratio check	XVIII	X	-∞	-40	-∞	-37.7	-∞	-37.7	-∞	-35.9	dB
X1, X2	Slew rate check	XIX	S	0.3	4.0	0.3	4.0	0.3	4.0	0.3	4.0	V/µs
Y	Frame continuity	XX	S	0	1	0	1	0	1	0	1	Ω

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater).

3/ "Q" parameters shall be tested during qualification testing only.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

* 4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

* 4.4.2 Order of testing. Test conditions A through Y shall be performed in any order with the following exceptions. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows: A, B where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

* 4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Y in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through U, W, and X shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 3. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25°C ±5°C ambient.

* 4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through U, W, and X shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests B through Y shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/202-1 supersedes part number M28787/202. The two parts are interchangeable.

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Supply voltage range	±9.5 V dc to ±16.5 V dc
Maximum power dissipation +16.5 V supply -16.5 V supply	 693 mW (no signal) 693 mW (no signal)
Maximum input voltage	±15 V or the V_{CC} applied, whichever is less
Input offset voltage	4.5 mV maximum
Input offset current	50 nA maximum
Input bias current	140 nA maximum
Input impedance	500 kΩ minimum
Common mode rejection ratio	-35.9 dB maximum
Supply voltage rejection ratio	-40 dB maximum
Slew rate (unit gain)	300 mV/µs minimum ($R_L \geq 49.9\Omega$, V_S = ±15 V)
Small signal bandwidth (unity gain)	430 kHz minimum (C1 disconnected)
Output voltage swing	±(V_{CC} - 4) (R_L = 49.9Ω)
Output drive current capability	200 mA minimum (see 6.5.1)
Large signal voltage gain	32 000 (V_{CC} = ±15 V)
Nominal gain options	-6.0 dB to +20.0 dB
Gain accuracy	±3.8 percent
Broadband self noise	200 µV rms maximum (input at ground, output open, gain = 11, 3 dB bandwidth 10 to 100 kHz)

6.5 Module function. This module contains one general purpose amplifier in series with a power driver amplifier.

* 6.5.1 Capacitive loading. Capacitor C1 (located within the module) can be connected in parallel with the feedback resistor to enable the module to drive a capacitive load. By connecting pin 14 to the output pin 16 and the feedback resistor, a capacitive load of up to 0.01 µF can be driven without the module output becoming unstable.

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* 6.5.2 Module maximum resistive loading. The output of the module is of a complementary emitter-follower design (two output transistors with one supplying positive voltage across the load and the other supplying negative voltage). If the module is operating with a dc signal voltage, only one of the output transistors is on, therefore, only one is dissipating power. With the power supply voltage at ±16.5 V dc and with a 50Ω module load, the maximum output voltage shall be limited to 4 V dc. Module output transistor power dissipation with any other combination of power supply voltage (V_{CC}), output voltage (V_{OUT}), and resistive load (R_L) shall be limited to 1 W in accordance with the following relationship:

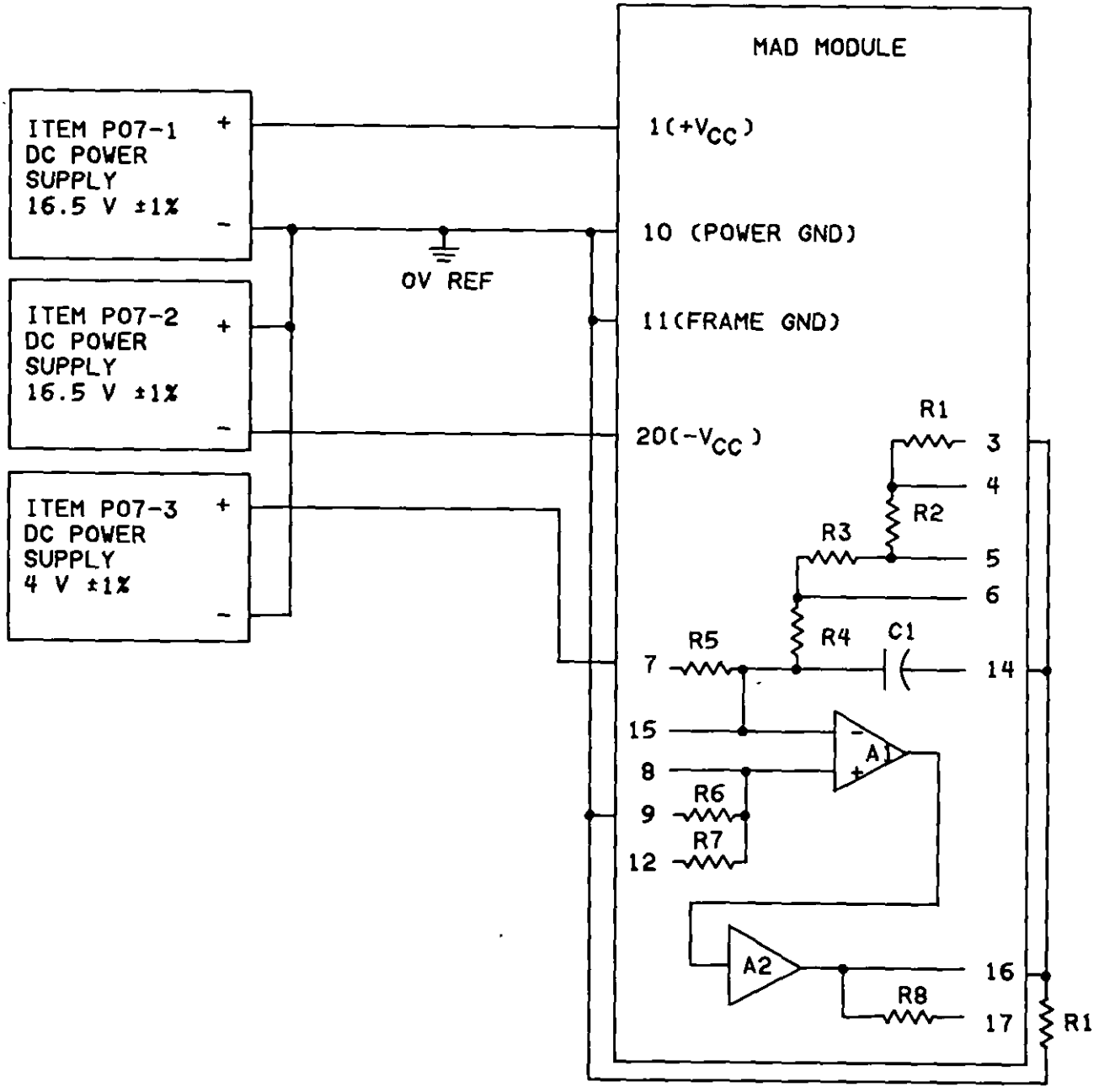
$$P = (V_{CC} - V_{OUT})\left(\frac{V_{OUT}}{R_L}\right) \leq 1 \text{ W}.$$

With an ac sine wave signal voltage, the two output transistors are alternately on and off at the frequency of the signal voltage, therefore, both are dissipating power. With the power supply voltage at ±16.5 V dc and with a 50Ω module load, the power dissipation will not be excessive for any sine wave voltage up to the maximum peak voltage allowed of 12.5 V (16.5 -4). If the load is less than 50Ω, power dissipation may be the limiting factor. Power dissipation of the two output transistors for a given power supply voltage (V_{CC}), output voltage (V_{pk}), and resistive load (R_L) shall be limited to 1.5 W total (0.75 W per output transistor) in accordance with the following relationship:

$$P_{TOTAL} = \frac{2}{\pi}\left(\frac{V_{CC}\ V_{pk}}{R_L}\right) - \left(\frac{V_{pk}}{2\ R_L}\right)^2 \leq 1.5 \text{ W}$$

Where V_{pk} = 1.414 V rms.

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NOTES:
1. R1 = 49.9Ω ±1%, 1/2 W.
2. The component temperature test shall be run with item P07-3 polarity as shown and with the polarity reversed.

FIGURE 2. Operating temperature, component temperature and life test circuit.

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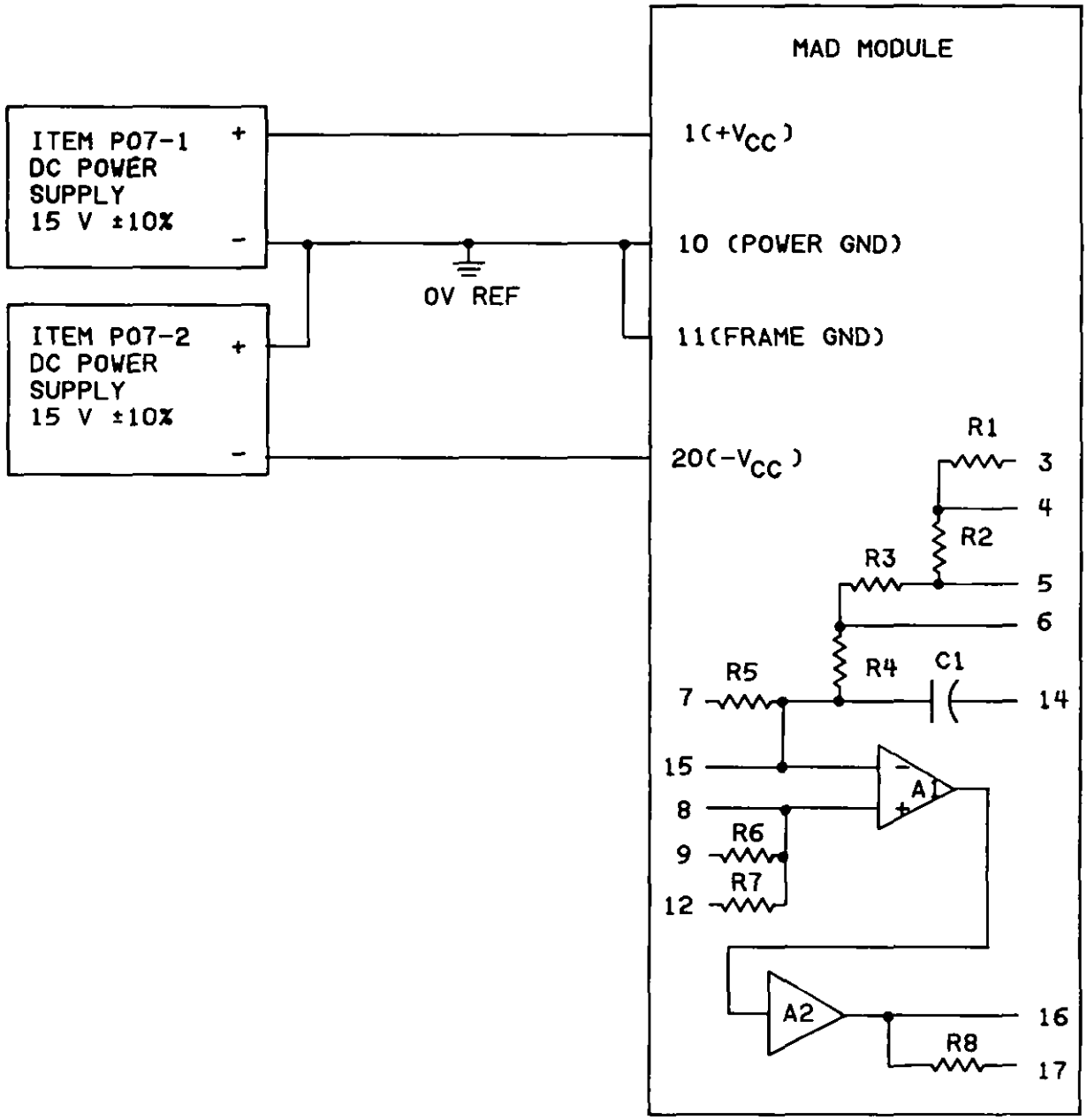


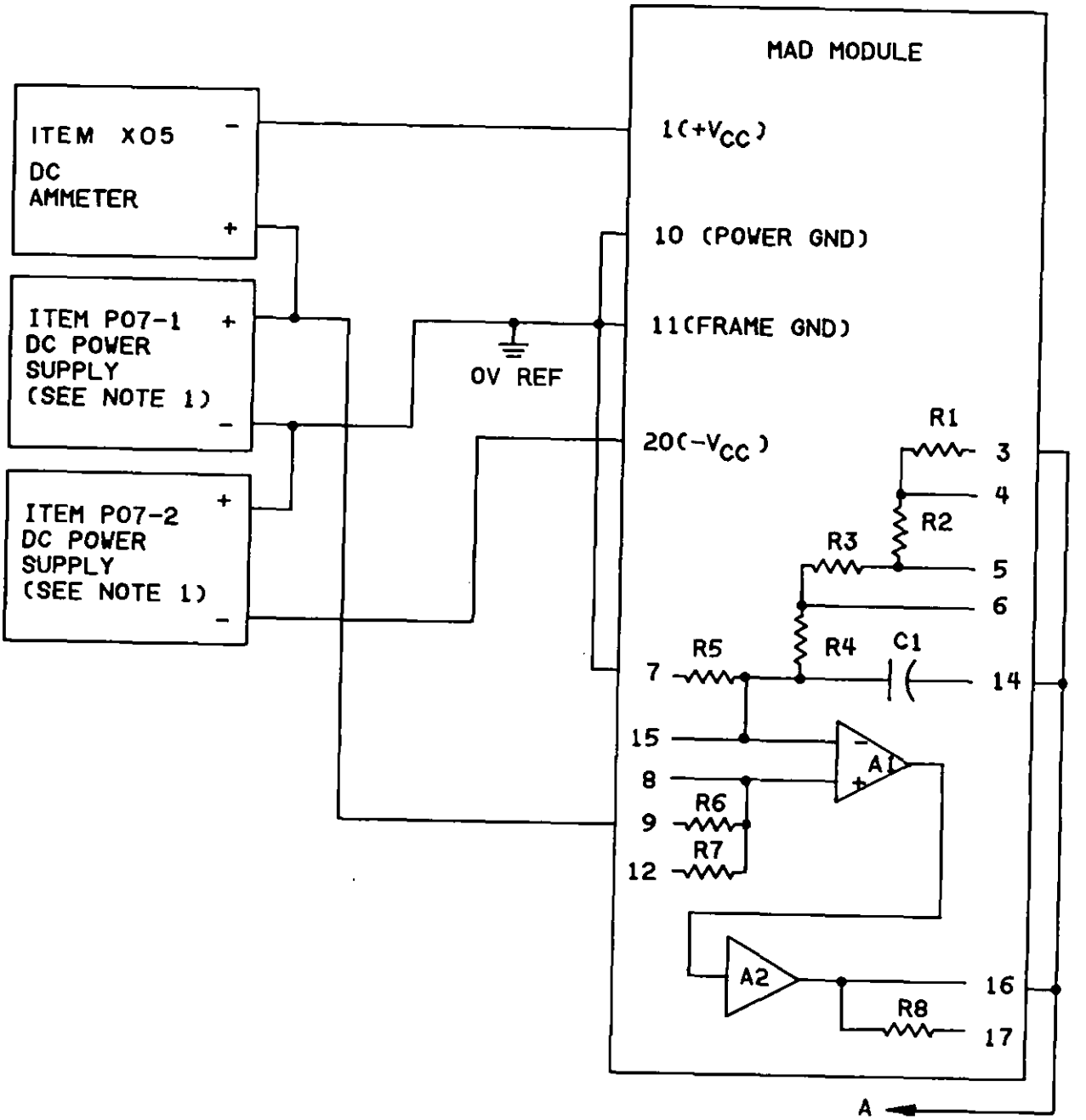
FIGURE 3. Vibration test circuit.

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* TABLE II. Output short circuit current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
A1	Output short circuit current (V_{CC} = ±10.5 V)	30	300	mA	4
A2	Output short circuit current (V_{CC} = ±16.5 V)	30	300	mA	4

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NOTES:
1. Adjust item PO7-1 and PO7-2 to 10.5 V ±1% for test condition A1 and to 16.5 V ±1% for test condition A2.
2. For test condition A1, the short circuit (point A to 0 V REF) shall be applied for 60 ±10 seconds before taking a reading.
3. For test condition A2, the short circuit (point A to 0 V REF) shall be applied for a maximum of 5 seconds.

FIGURE 4. Output short circuit current test circuits for table II.

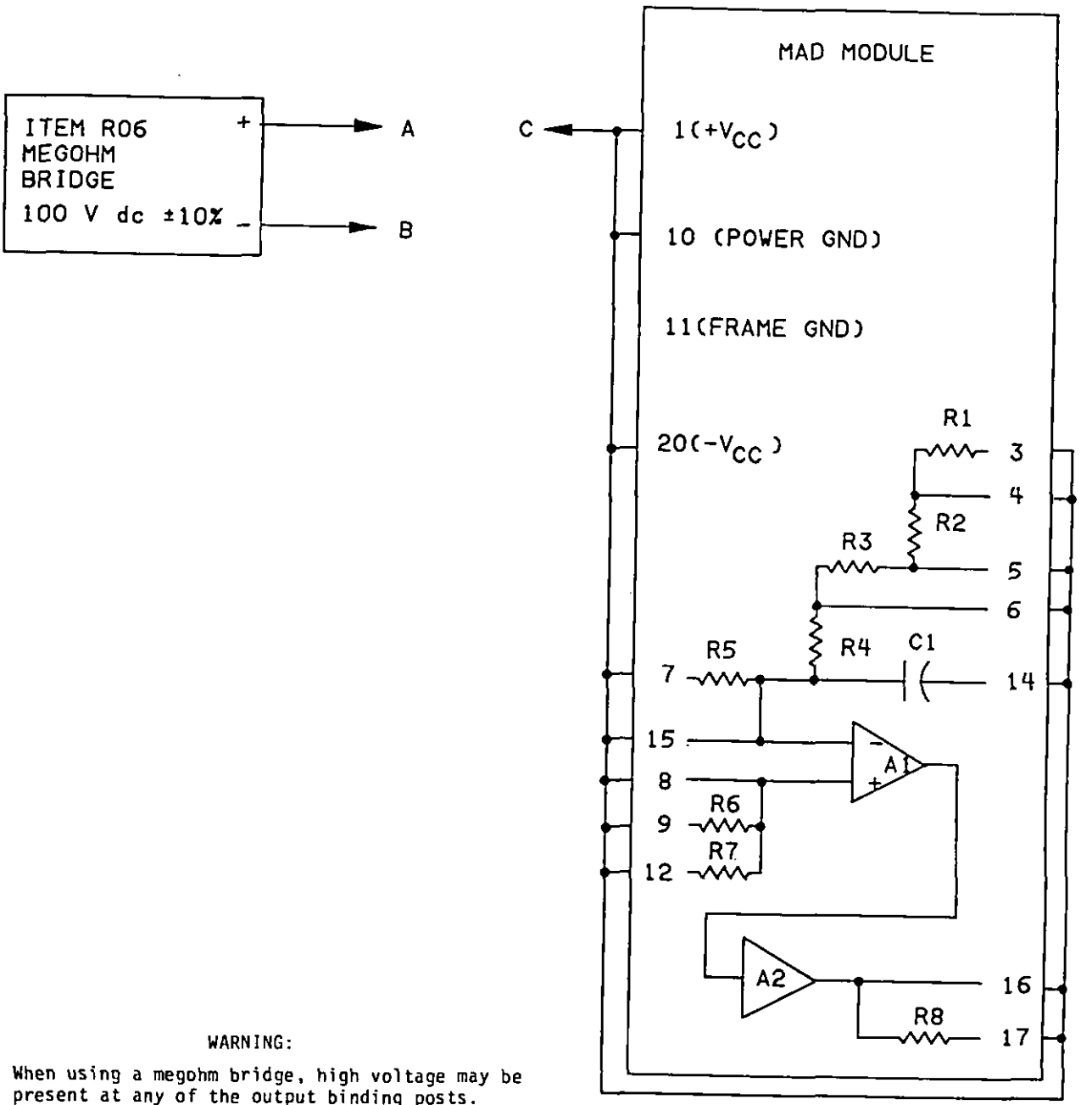
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TABLE III. Isolation test. 1/

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Isolation, frame to module pins	C	11	10	∞	MΩ	5
B2	Isolation, used to unused pins	C	2, 13, 18, 19, 21-40	"	"	"	"
B3	Isolation, unused to unused pins	2	13, 18, 19, 21-40	"	"	"	"
34	Isolation, unused to unused pins	13	18, 19, 21-40	"	"	"	"
B5	Isolation, unused to unused pins	18	19, 21-40	"	"	"	"

1/ For test conditions B2-B5, the optional pins 21-40 may not be on the module. When the module does include pins 21-40, connect them as specified above.

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

FIGURE 5. Isolation test circuit for table III.

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TABLE IV. Power supply current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
C1	Power supply current (16.5 V)	1	39	mA	6
C2	- Power supply current (16.5 V)	1	39	mA	6

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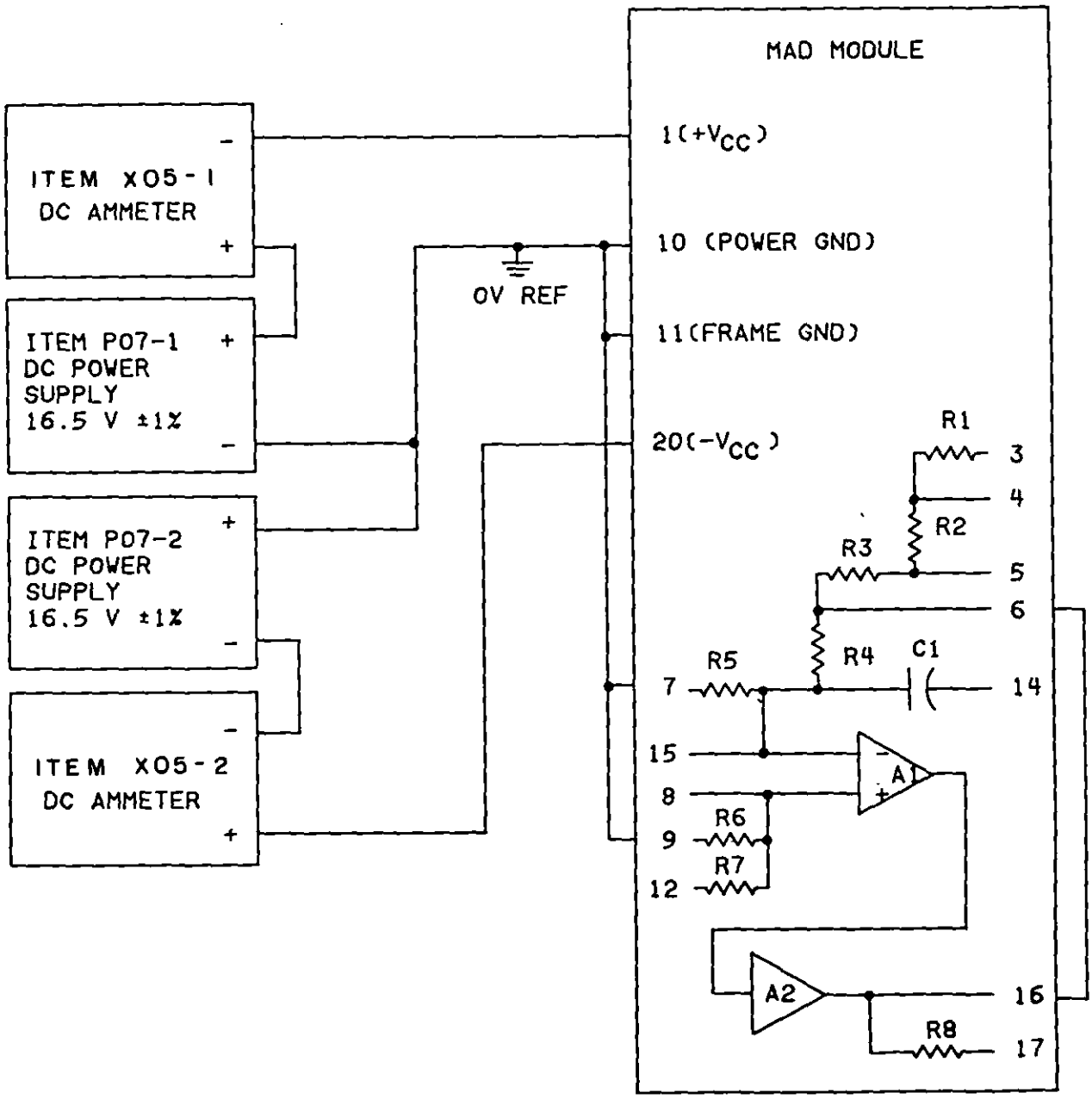


FIGURE 6. Power supply current test circuit for table IV.

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TABLE V. Input bias current check test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
D	Input bias current check 1/	-0.1	-28	mV	7

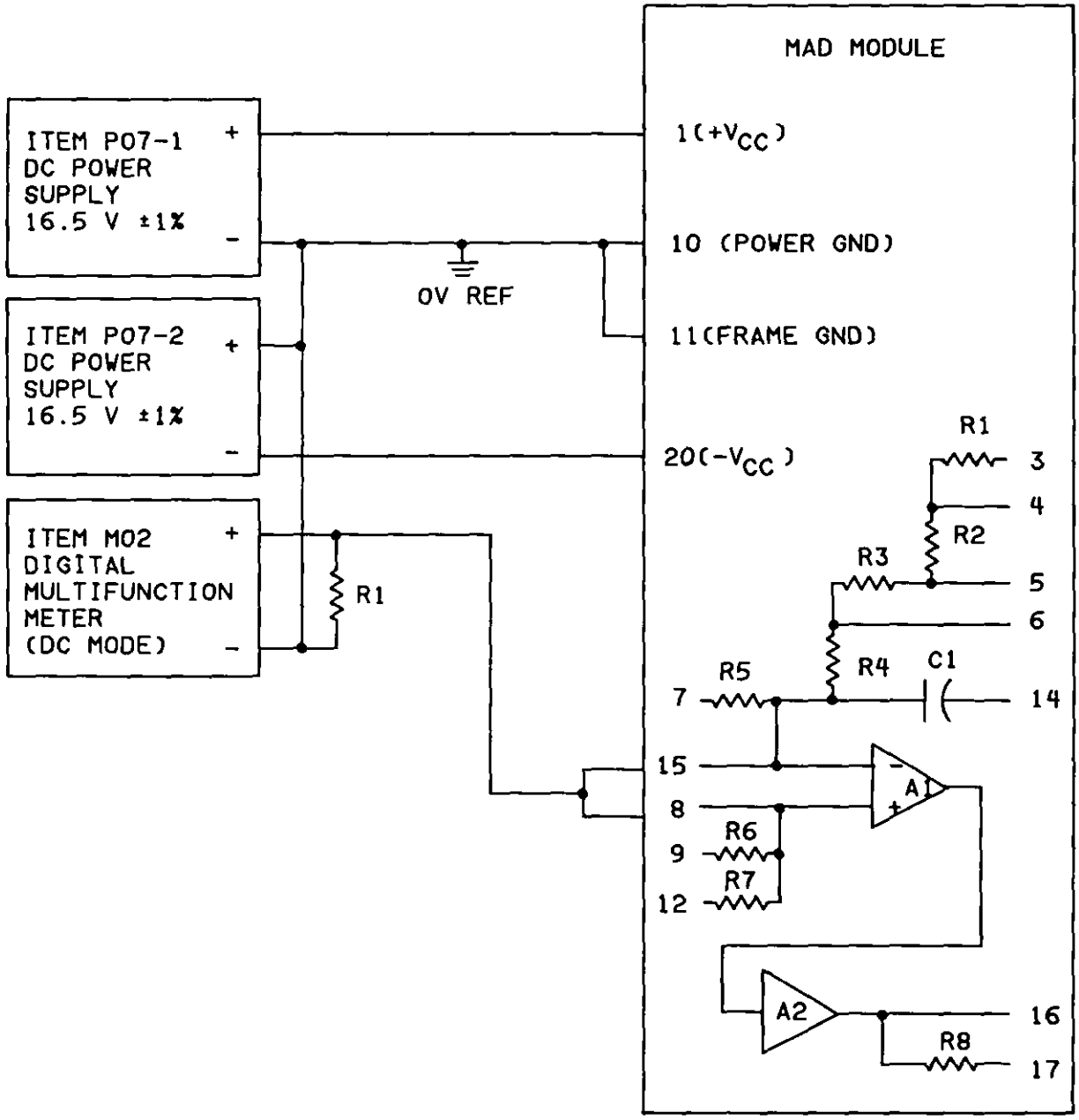
1/ Test condition D verifies an input bias current per input of 140 nA in accordance with the following relationship:

$$\text{Input bias current (A each input)} = \frac{1}{2}\left(\frac{|V_{R1}|}{100\ k\Omega}\right)$$

where:

V_{R1} is the voltage across R1 in volts.

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NOTE: R1 = 100 kΩ ±1%, 1/8 W.

* FIGURE 7. Input bias current check test circuit for table V.

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TABLE VI. Input offset voltage and input offset current check test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
E1	Input offset voltage (R source = 50Ω) check	-350	350	mV	8
E2	Input offset voltage (R source = 8.25 kΩ) check	-350	350	"	8
F	Input offset current (F = E2 - E1) check	-41.2	41.2	"	-

NOTES:

1. Test conditions E1 and E2 verify an input offset voltage of ±3.5 mV maximum (±4.5 mV maximum over the operating temperature range) in accordance with the following relationship:

$$\text{Input offset voltage (mV)} = \frac{\text{E1 or E2 value in mV}}{\text{Gain of 101}}$$

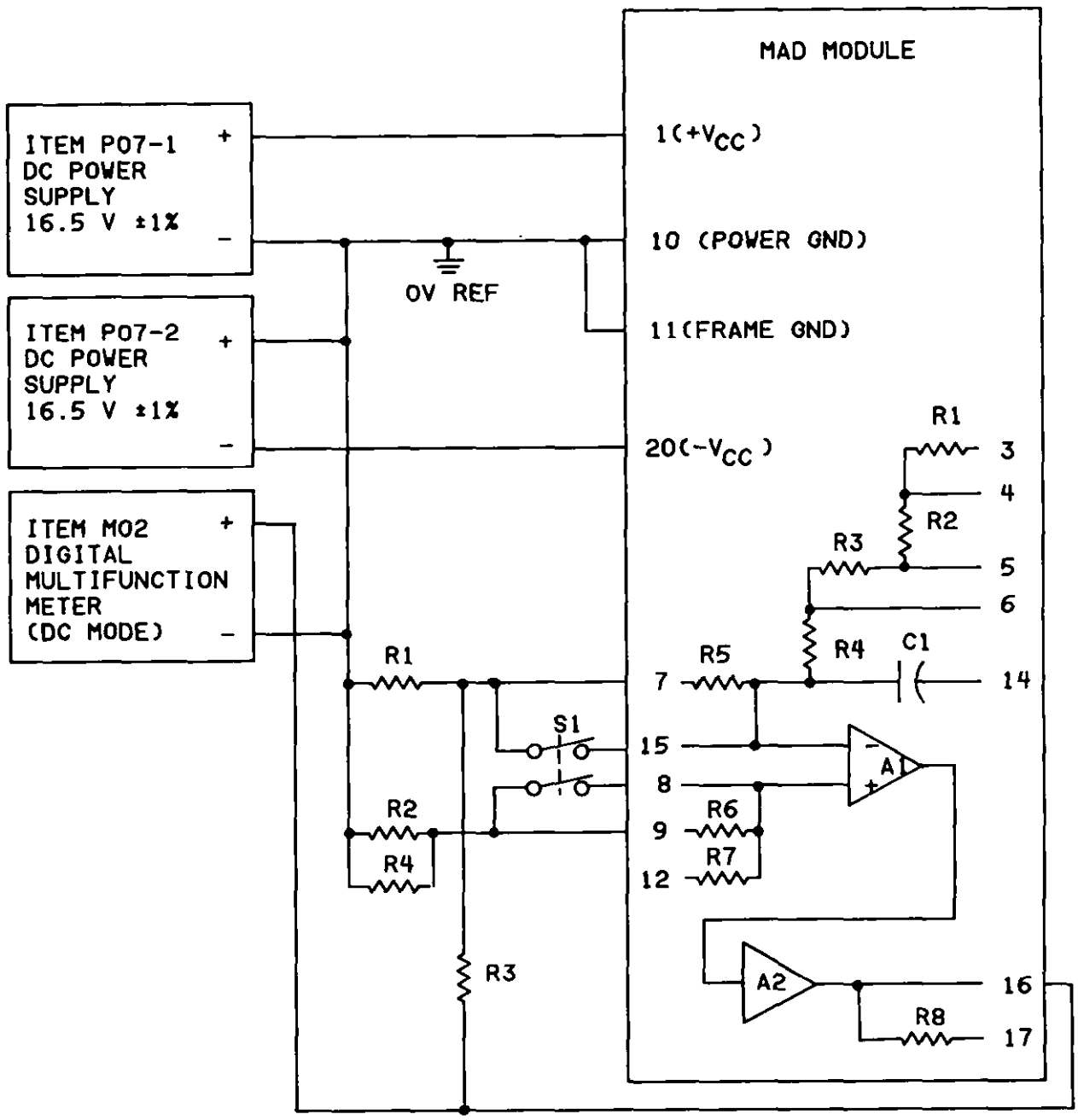
2. Test condition F is a calculation. For test condition F, subtract the voltage measured in test condition E2 from that measured in test condition E1. Test condition F verifies an input offset current of ±50 nA in accordance with the following relationship:

$$\text{Input offset current (A)} = \frac{\text{E2 value - E1 value}}{(8\ 250\Omega)(\text{Gain of 101})}$$

where:

E1 and E2 are in volts.

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NOTES:
1. R1 = R2 = 49.9Ω ±0.1%, 1/8 W.
2. R3 = R4 = 4.99 kΩ ±0.1%, 1/8 W.
3. S1 = DPST switch.
4. Close switch S1 for test condition E1 and open switch S1 for test condition E2.

* FIGURE 8. Input offset voltage and input offset current check test circuit for table VI.

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TABLE VII. Inverting gain check test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
G1	Inverting gain check (V_{CC} = ±10 V)	3	0.975	1.035	V rms	9
G2	"	4	0.3880	0.4120	"	"
G3	"	5	0.1940	0.2060	"	"
G4	"	6	0.0970	0.1030	"	"
G5	Inverting gain check (V_{CC} = ±15 V)	3	0.975	1.035	"	"
G6	"	4	0.3880	0.4120	"	"
G7	"	5	0.1940	0.2060	"	"
G8	"	6	0.0970	0.1030	"	"

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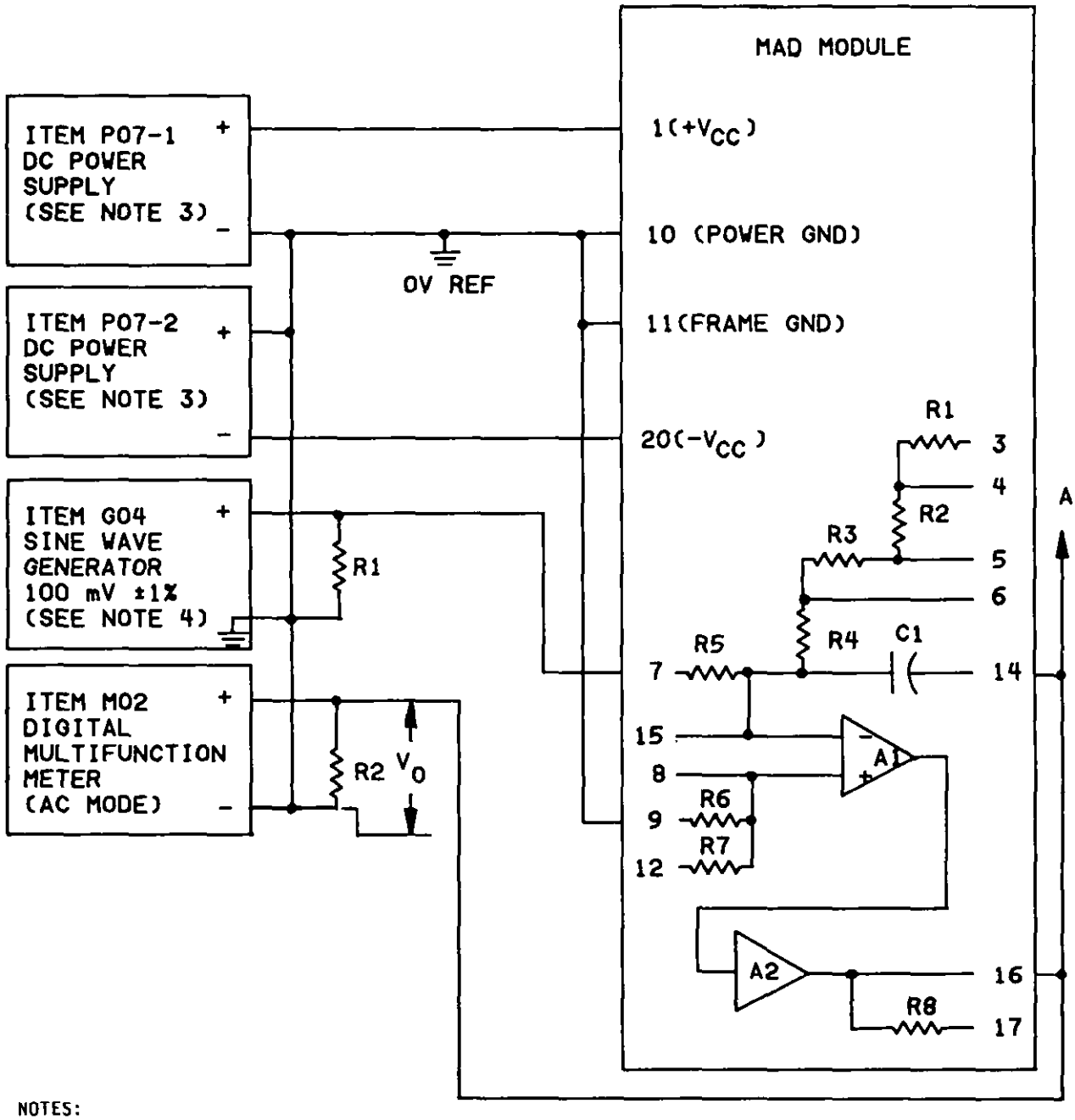
TABLE VIII. Inverting frequency response check test. 1/

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
H	Inverting frequency response check (C1 connected)	3	630	760	mV rms	9

1/ Test condition H verifies that the frequency response is -3.00 +0.6, -1.0 dB (-3.00 +0.8, -1.2 dB at 25°C ambient end-of-life and initial 60°C operating temperature and -3.00 +1.0, -1.4 dB at 60°C operating temperature end-of-life) down at 12.8 kHz ±2% in accordance with the following relationship:

$$dB = 20 \log \left(\frac{V_O}{1\ V} \right).$$

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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. R2 = 49.9Ω ±1%, 1/8 W.
3. Adjust items P07-1 and P07-2 to 10 V ±1% for test conditions G1-G4 and to 15 V ±1% for test conditions G5-G8 and H.
4. Adjust item G04 frequency to 1 kHz ±2% for test conditions G1-G8 and to 12.8 kHz ±2% for test condition H.
5. Stray capacitance between pins 14 and 15 shall be kept to a minimum. Stray capacitance will affect the readings.

* FIGURE 9. Inverting gain check and inverting frequency response check test circuit for tables VII and VIII.

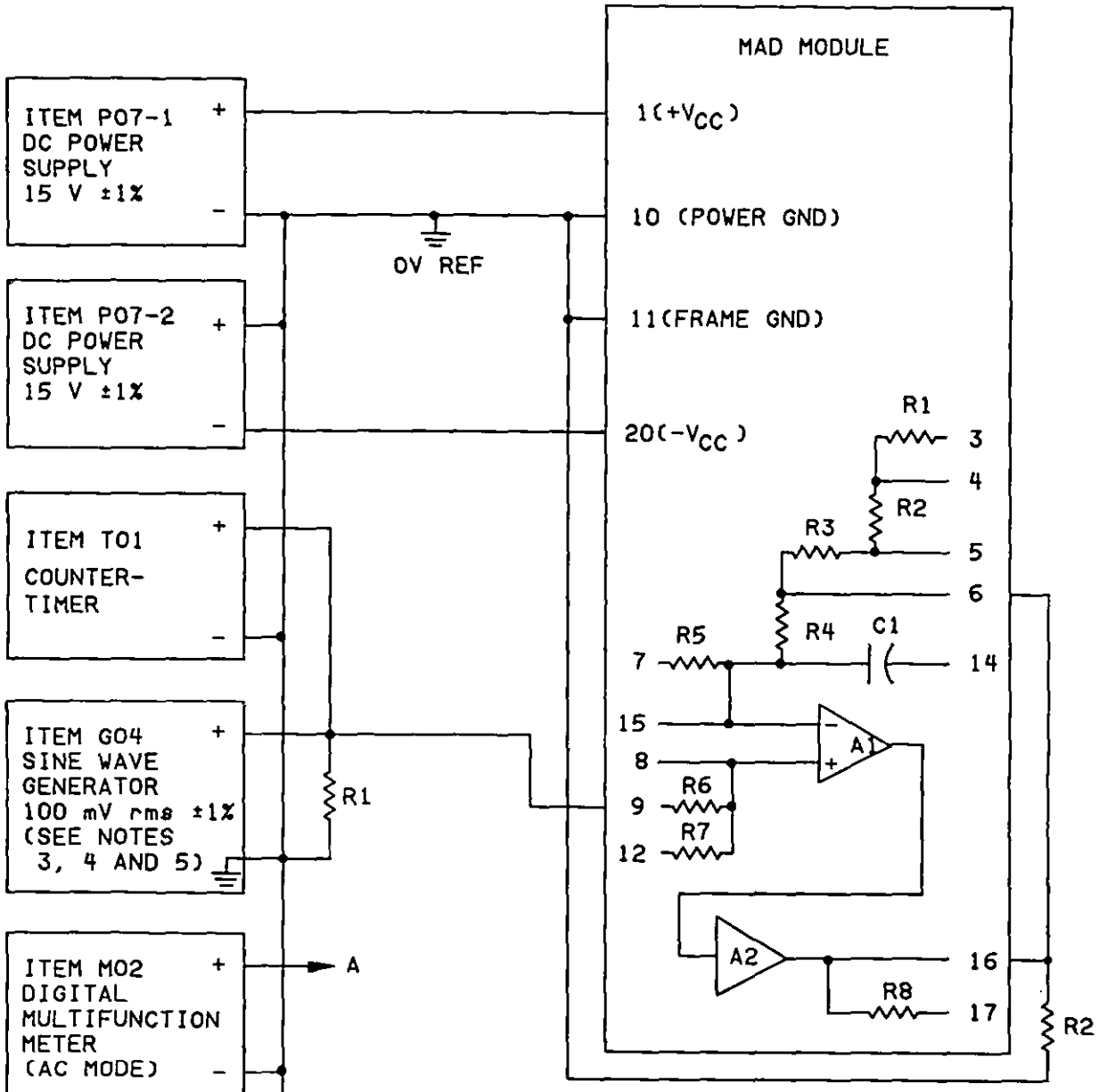
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* TABLE IX. Noninverting gain check, noninverting phase margin check, and noninverting bandwidth test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J	Noninverting gain check	6, 16	Open	98	102	mV rms	10
K	Noninverting phase margin check (gain peaking) 1/	Open	6, 16	98	141	mV rms	"
L	Noninverting bandwidth (C1 disconnected)	Open	6, 16	0.43	3.0	MHz	"

1/ Test condition K verifies a noninverting phase margin of 45° minimum by checking the gain peaking to be +3 dB (1.41 V/V) maximum.

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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. R2 = 49.9Ω ±1%, 1/8 W.
3. For test condition J, adjust item G04 frequency to 1 kHz ±2%.
4. For test condition K, increase the frequency of item G04 until the output voltage peaks at a maximum value. Check the module input voltage when the peak is found to insure that it is still within the required limits. The peak voltage shall meet the requirements for test condition K. Use item W04 to measure the input and output voltage.
5. For test condition L, increase the frequency beyond the peak voltage until the output voltage is 71 mV. Check the module input voltage to insure that it is still within the required limits. The item G04 frequency at this output voltage shall meet the requirements of the applicable test condition. Use item W04 to measure the input and output voltage.

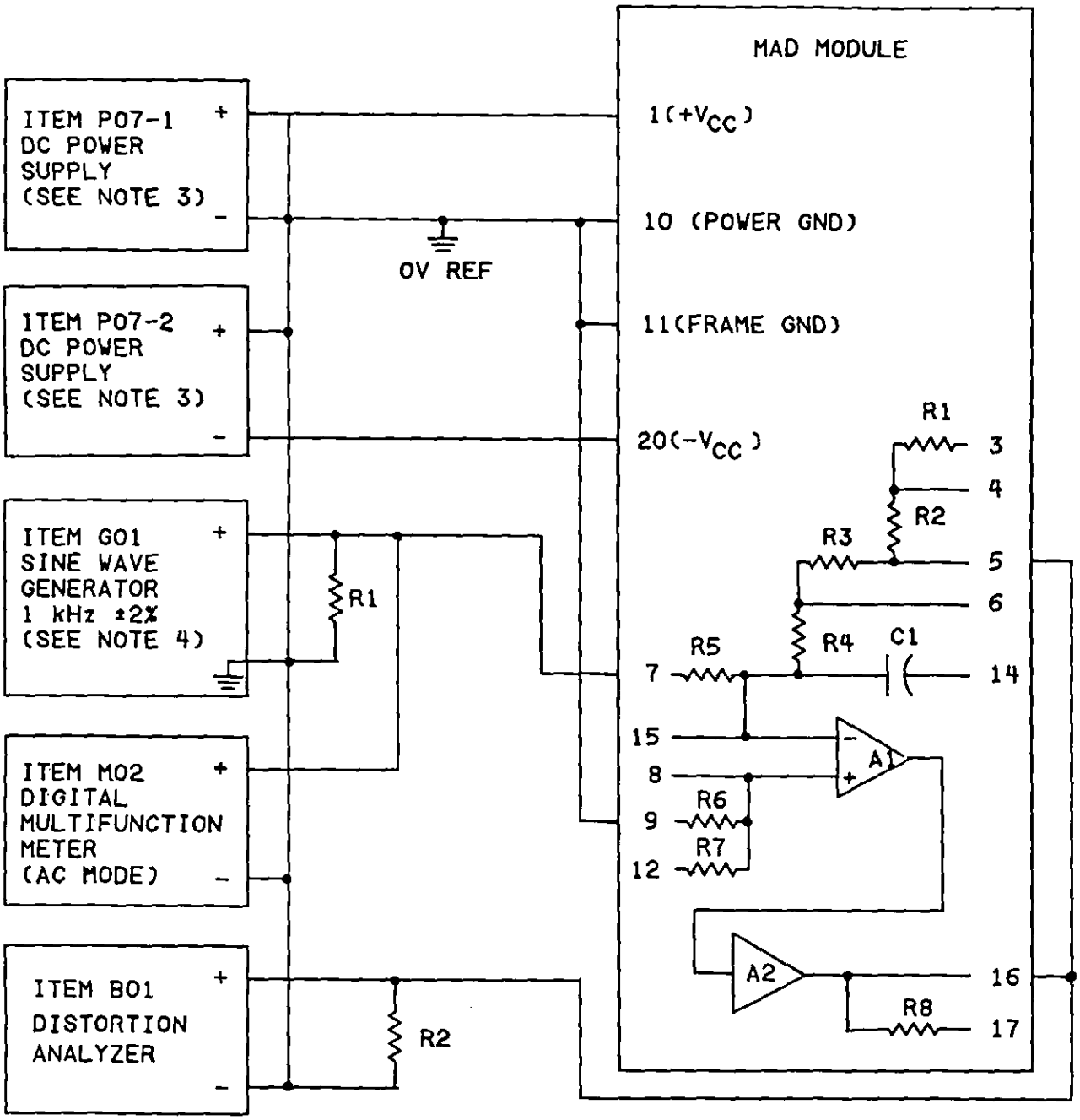
* FIGURE 10. Noninverting gain check, noninverting phase margin check, and noninverting bandwidth test circuit for table IX.

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TABLE X. Distortion test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
M1	Distortion (V_{CC} = 9.5 V)	0	1	%	11
M2	Distortion (V_{CC} = 13.5 V)	0	1	%	11

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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. R2 = 49.9Ω ±1%, 1/8 W.
3. Set items P07-1 and P07-2 to 9.5 V ±1% for test condition M1 and to 13.5 V ±1% for test condition M2.
4. Set item G01 voltage to 1.940 V rms ±1% for test condition M1 and to 3.350 V rms ±1% for test condition M2.
5. This test also verifies the output voltage swing to be ±9.5 V_{pk} with $\pm V_{CC}$ at 13.5 V and ±5.5 V_{pk} with V_{CC} at 9.5 V.
6. For each test condition, subtract the input distortion from the output distortion.

FIGURE 11. Distoration test circuit for table X.

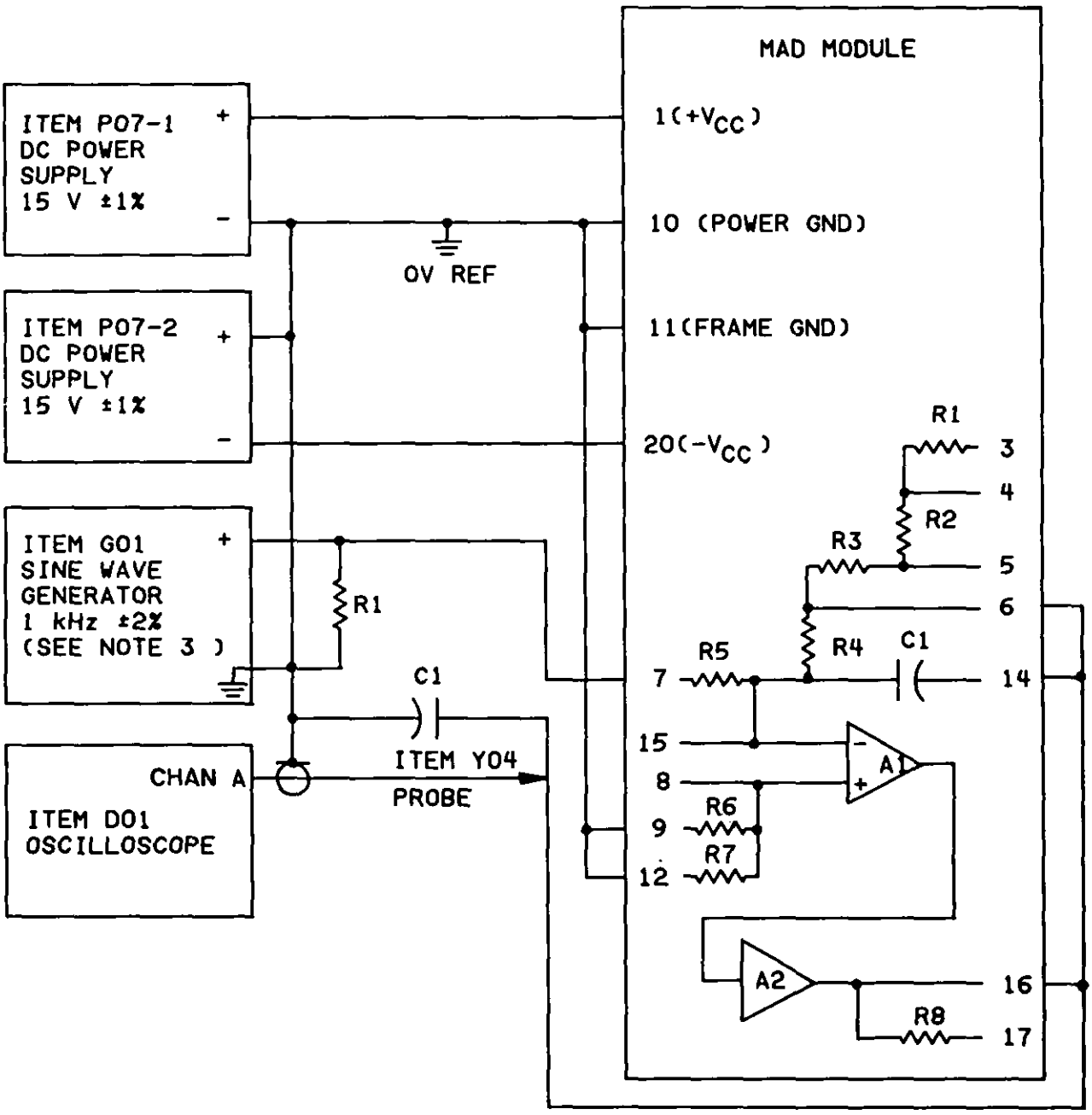
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TABLE XI. Oscillation check test. 1/

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
N1	Oscillation check (capacitive load)	0	5	mVp-p	12
N2	Oscillation check (capacitive load)	"	600	"	12
N3	Oscillation check (cable load)	"	5	"	13
N4	Oscillation check (cable load)	"	600	"	13

1/ Oscillation is defined as any signal on the output at a frequency other than the 1 kHz being input (usually the oscillation will be much greater than 1 kHz).

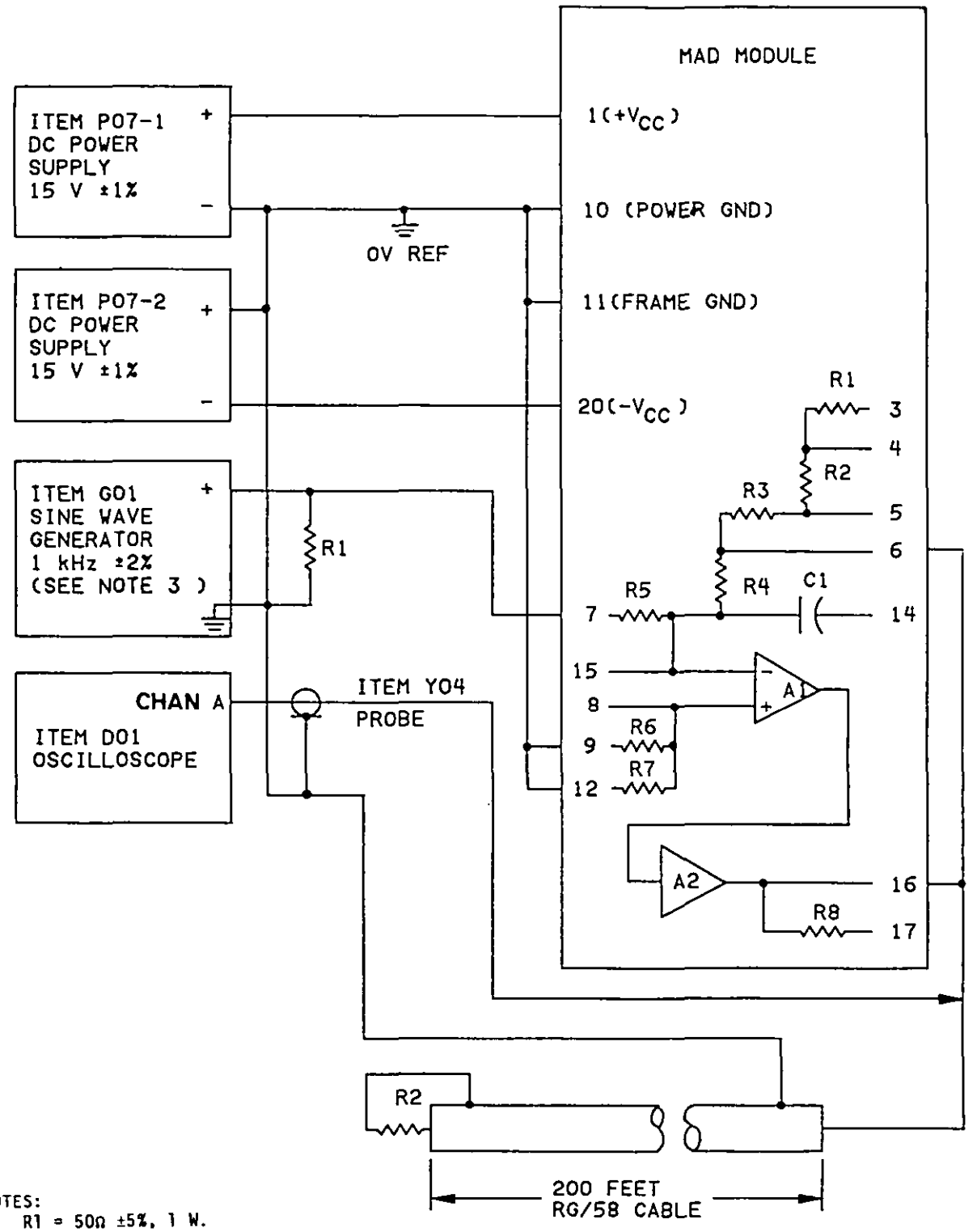
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NOTES:
1. R1 = 600Ω ±5%, 1/4 W.
2. C1 = 0.01 µF ±5%, 300 V dc, glass dielectric.
3. Set item G01 voltage to 100 mV rms ±1% for test condition N1 and to 7.5 V rms ±1% for test condition N2.
4. Vary the item D01 horizontal sweep rate to check for oscillations. Any oscillations shall meet the limits specified in table XI.

FIGURE 12. Oscillation check test circuit for test conditions N1 and N2, table XI.

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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. R2 = 50Ω ±5%, 1.5 W.
3. Set item G01 voltage to 100 mV rms ±1% for test condition N3 and to 7.5 V rms ±1% for test condition N4.
4. Vary the item D01 horizontal sweep rate to check for oscillations. Any oscillations shall meet the limits specified in table XI.

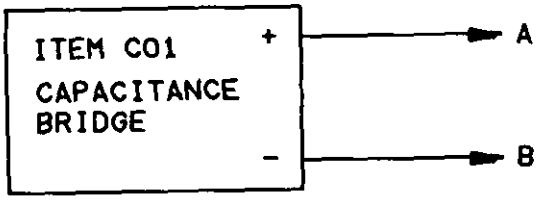
FIGURE 13. Oscillation check test circuit for test conditions N3 and N4, table XI.

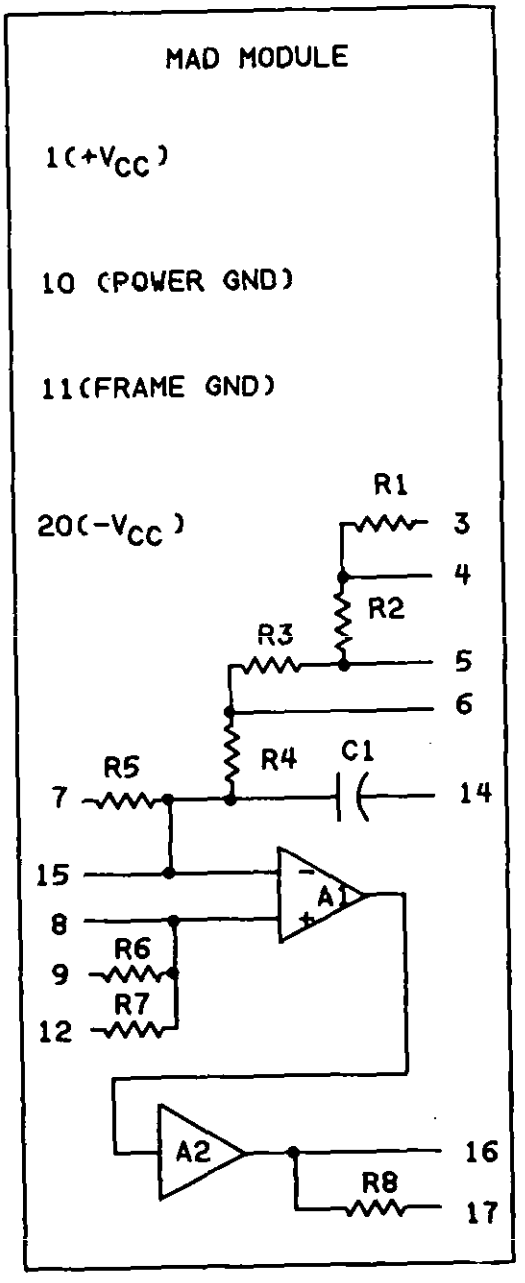
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TABLE XII. Capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
P1	Power supply capacitance (+)	1	10	0.75	1.25	µF	14
P2	Power supply capacitance (-)	10	20	0.75	1.25	µF	14

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NOTE: Item C01 setup:
Frequency = 120 Hz ±5%.
V_{pk} = 300 mV maximum.

* FIGURE 14. Capacitance test circuit for table XII.

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TABLE XIII. Resistance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
R1	Resistance (R1)	3	4	49.40	50.40	kΩ	15
R2	Resistance (R2)	5	4	16.33	16.67	"	"
R3	Resistance (R3)	5	6	8.167	8.333	"	"
R4	Resistance (R4)	15	6	"	"	"	"
R5	Resistance (R5)	15	7	"	"	"	"
R6	Resistance (R6)	8	9	"	"	"	"
R7	Resistance (R7)	8	12	"	"	"	"
R8	Resistance (R8)	16	17	48.45	53.55	Ω	"

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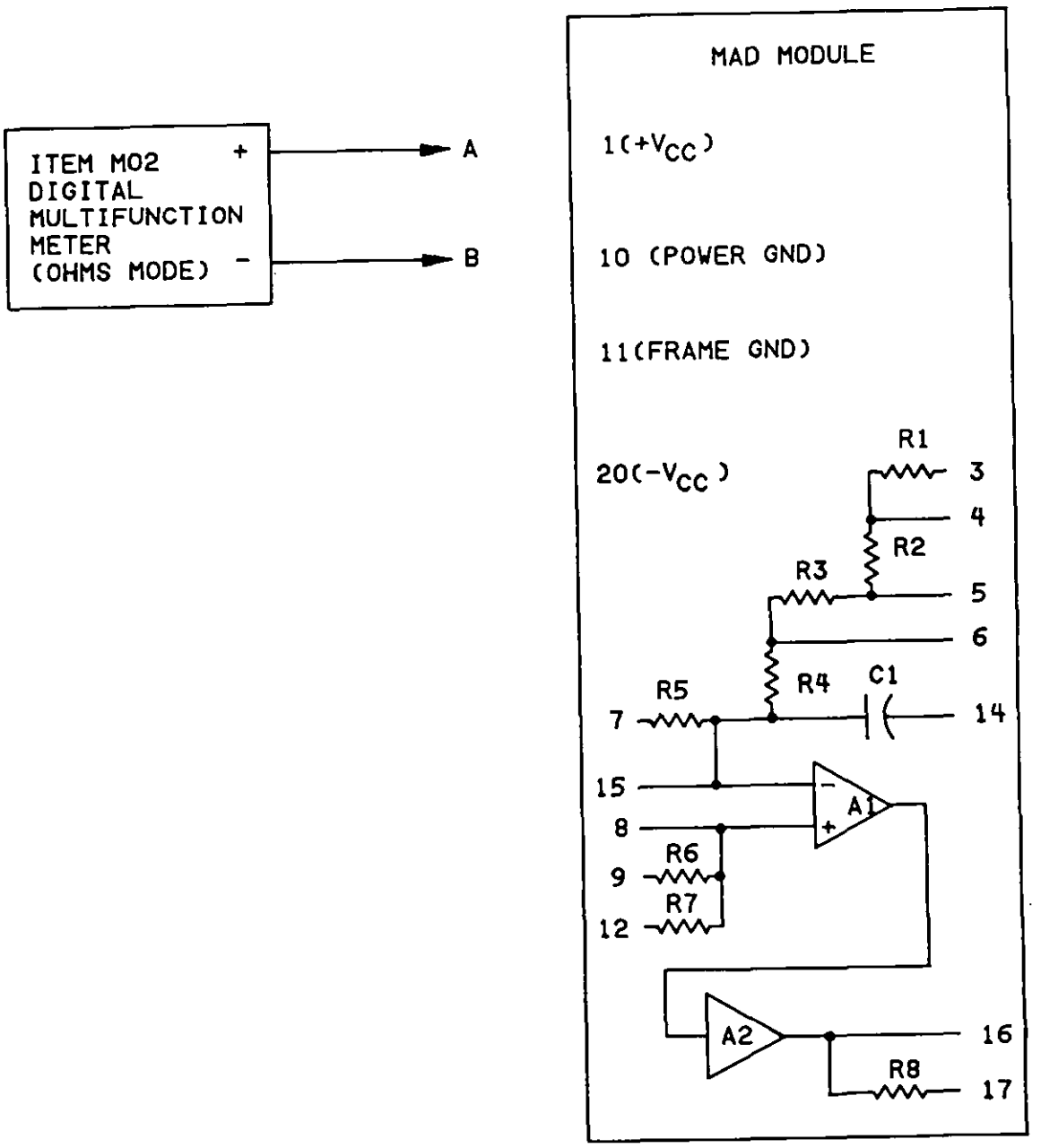


FIGURE 15. Resistance test circuit for table XIII.

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TABLE XIV. Large signal voltage gain check test. 1/

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
S1	Large signal voltage gain check	1	20	mV	16
S2	Large signal voltage gain check	1	8	mV	16

1/ This test verifies a large signal voltage gain of 20 000 minimum for test condition S1 and and a gain of 50 000 minimum (32 000 minimum over the operating temperature range) for test condition S2 in accordance with the following relationship:

$$A_{VS} = 100\left(\frac{V_O}{V_{IN}}\right) = 100\left(\frac{4\text{ V rms}}{V_{IN}}\right)$$

where:

V_O is the voltage set at pin 16 (4 V rms) and V_{IN} is the voltage measured for test condition S1 or S2.

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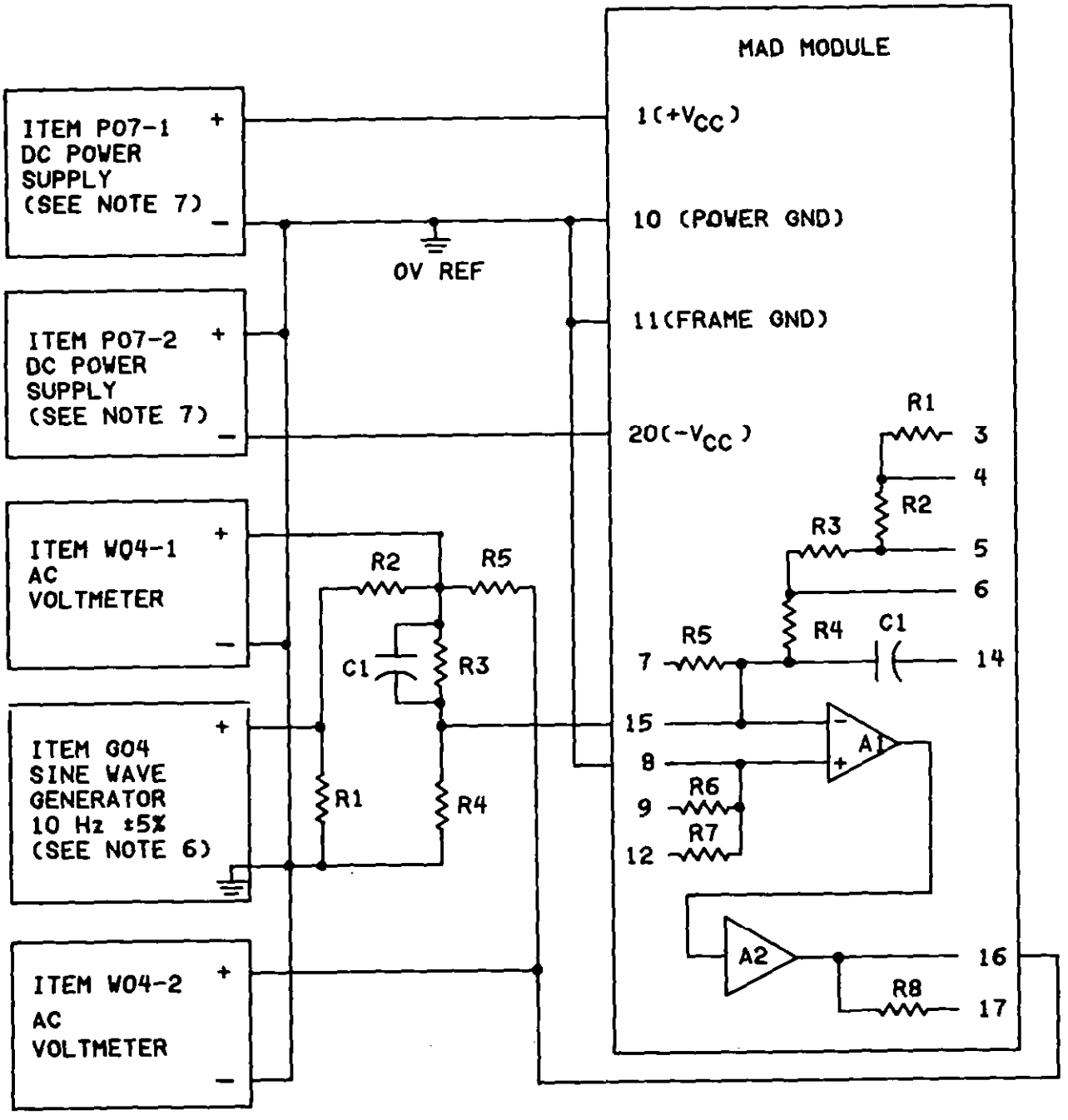


FIGURE 16. Large signal voltage gain check test circuit for table XIV.

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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. R2 = R5 = 49.9 kΩ ±1%, 1/8 W.
3. R3 = 100 kΩ ±1%, 1/8 W.
4. R4 = 1 kΩ ±1%, 1/8 W.
5. C1 = 100 pF ±10%, 50 V dc.
6. The test shall be performed as follows for each test condition:

   a. Connect the module.
   b. Adjust item G04 amplitude until item W04-2 reads 4 V rms ±1%.
   c. Read item W04-1 voltage. This voltage shall meet the requirements of table XIV.

7. Set items P07-1 and P07-2 to 10 V ±1% for test condition S1 and to 15 V ±1% for test condition S2.

FIGURE 16. Large signal voltage gain check test circuit for table XIV - Continued.

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TABLE XV. Input impedance check test. 1/

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
T	Input impedance check	2.86	4.04	V rms	17

1/ This test verifies a differential input impedance (Z_{id}) of 1 MΩ minimum (500 kΩ minimum over operating temperature range) in accordance with the following relationship:

$$Z_{id} = \frac{V_{02}\ (R6+R7)}{V_{01} - V_{02}} = \frac{V_{02}\ (4 \times 10^5 \Omega)}{4.0 - V_{02}}$$

where:

V_{01} is the voltage at pin 16 with S1 closed (4 V) and V_{02} is the voltage at pin 16 with S1 open.

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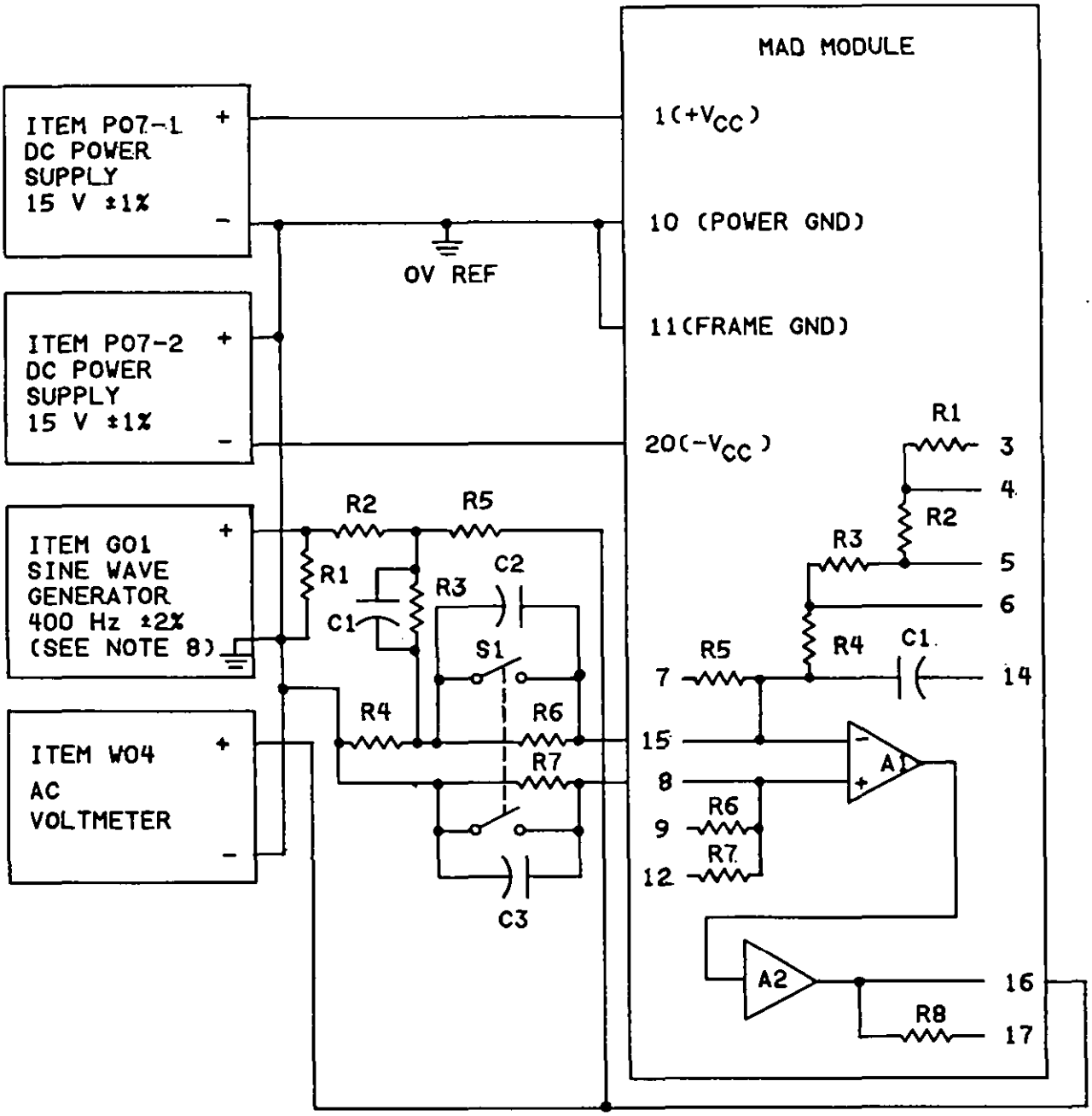


FIGURE 17. Input impedance check test circuit for table XV.

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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. R2 = R5 = 49.9 kΩ ±1%, 1/8 W.
3. R3 = 100 kΩ ±1%, 1/8 W.
4. R4 = 1 kΩ ±1%, 1/8 W.
5. R6 = R7 = 200 kΩ ±1%, 1/8 W.
6. C1-C3 = 100 pF ±10%, 50 V dc.
7. S1 = DPST switch.
8. The test shall be performed as follows:

   a. Connect the module.
   b. Close S1 and adjust item G01 until the pin 16 voltage as read on item W04 is 4 V rms ±1%.
   c. Open S1 and read the item W04 voltage. This voltage shall meet the requirements of table XV.

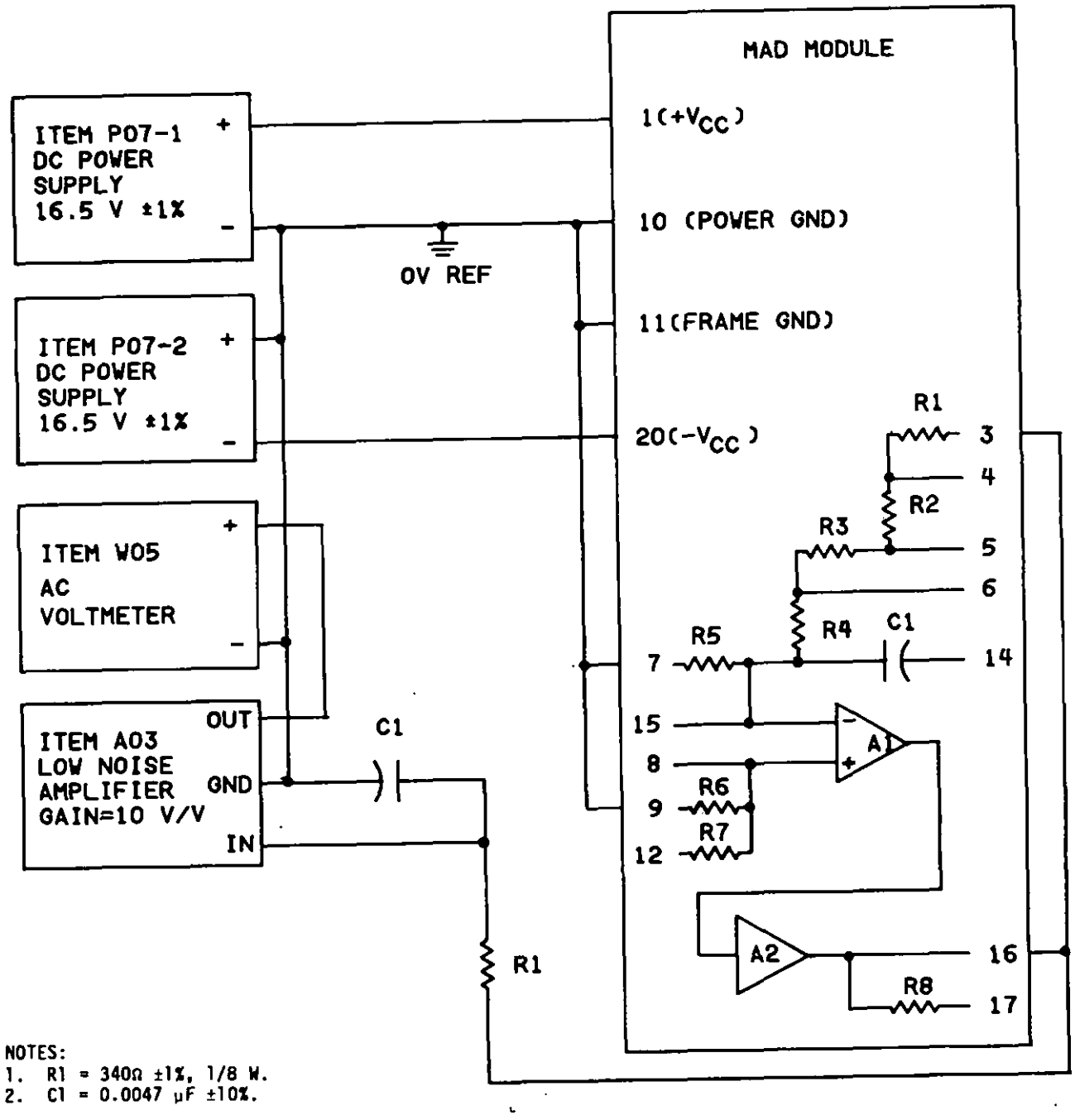
FIGURE 17. Input impedance check test circuit for table XV - Continued.

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* TABLE XVI. Broadband noise test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
U	Broadband noise	0	2	mV rms	18

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NOTES:
1. R1 = 340Ω ±1%, 1/8 W.
2. C1 = 0.0047 µF ±10%.

FIGURE 18. Broadband noise test circuit for table XVI.

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TABLE XVII. Supply voltage rejection ratio check test. 1/

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
V	Supply voltage rejection ratio check (1 kHz)	0	100	mV rms	19

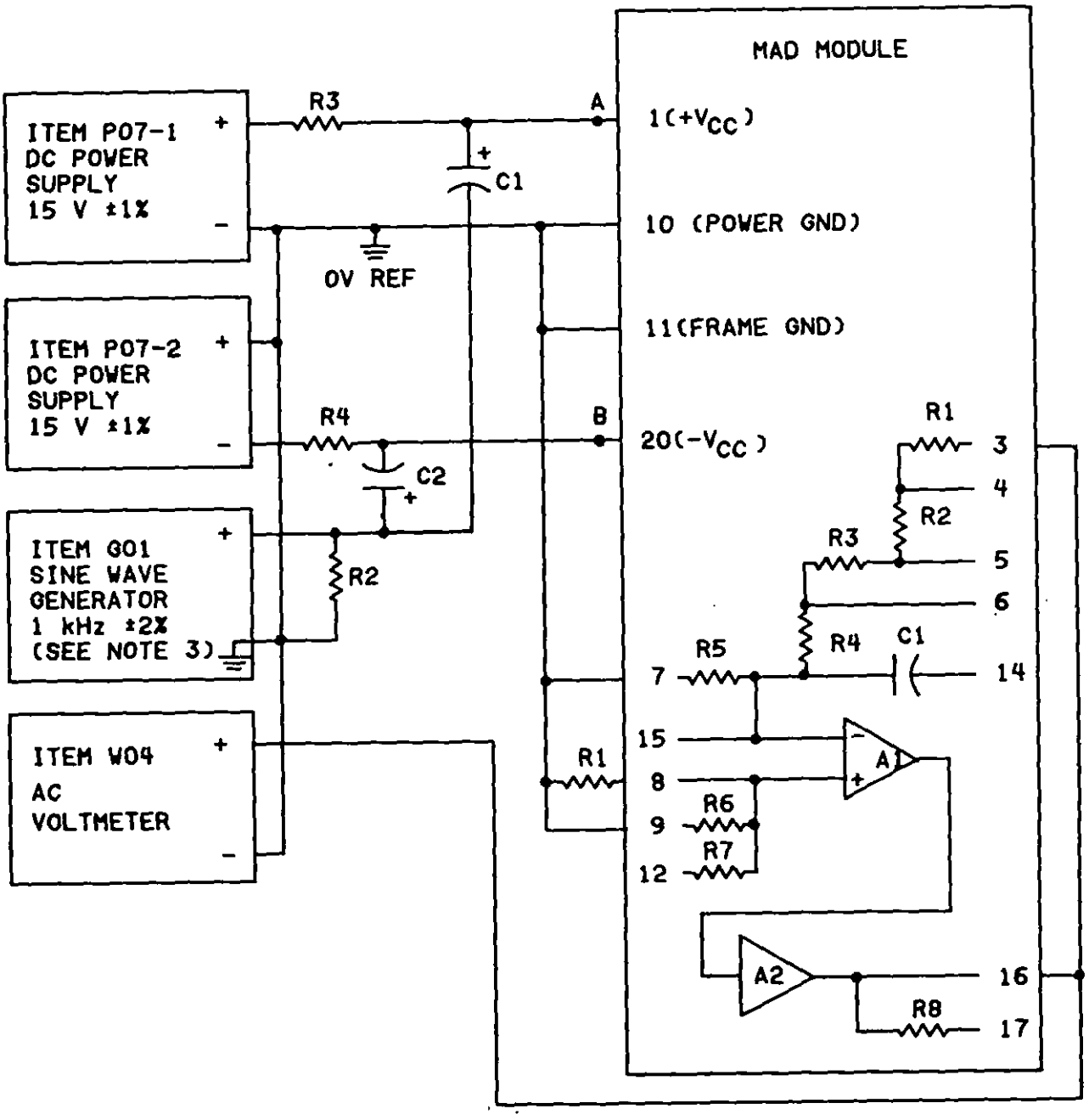
1/ This test verifies a supply voltage rejection ratio of less than -40 dB in accordance with the following relationship:

$$\text{Rejection ratio} = 20 \log \left(\frac{\frac{V_{out}}{Gain}}{V_{supply}} \right) = 20 \log \left(\frac{V_{out}}{10} \right)$$

where:

V_{supply} is the rms voltage at points A or B (1 V rms).
V_{out} is the voltage at pin 16 in volts rms and gain is the module voltage gain of 10.

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NOTES:
1. R1 = 82.5 kΩ ±1%, 1/8 W.
2. R2 = 50Ω ±5%, 1 W.
3. R3 = R4 = 47Ω ±5%, 1/4 W.
4. C1 = C2 = 20 µF ±20%, 20 V dc.
5. Adjust item G01 voltage for 1 V rms ±5% at points A and B.

FIGURE 19. Supply voltage rejection ratio check test circuit for table XVII.

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TABLE XVIII. Common mode rejection ratio check test. 1/

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
W	Common mode rejection ratio check (1 kHz)	0	75	mV rms	20

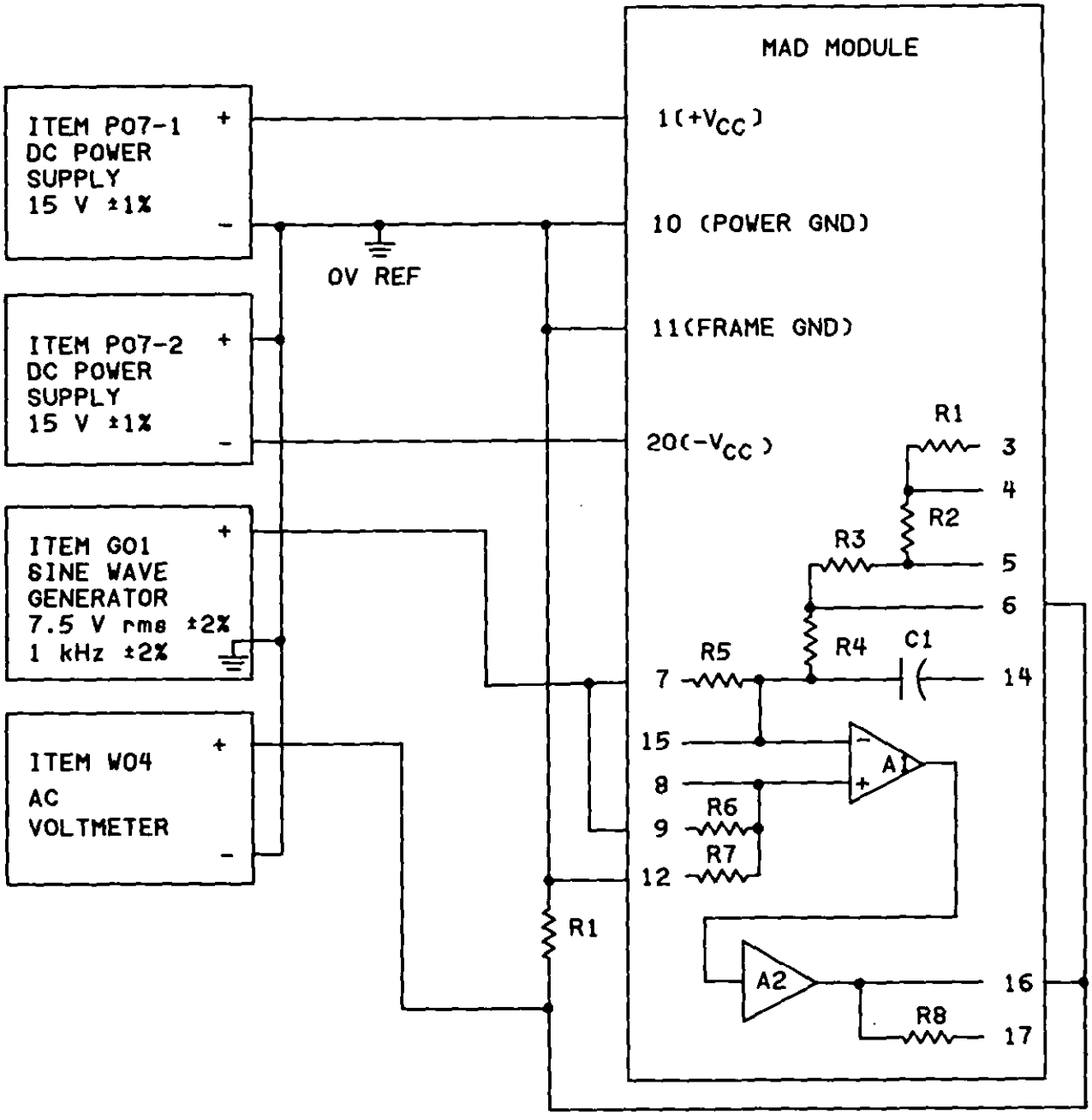
1/ This test verifies a common mode rejection ratio of less than -40 dB (-37.7 dB at 25° ambient end-of-life and 60°C initial operating temperature and -35.9 dB at 60°C operating temperature end-of-life) in accordance with the following relationship:

$$CMRR = 20 \log \frac{V_{out}}{V_{CM}} = 20 \log \frac{V_{out}}{7.5 \text{ V rms}}$$

where:

V_{out} is the output voltage at pin 16 in V rms.

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NOTE: R1 = 49.9Ω ±1%, 1/8 W.

FIGURE 20. Common mode rejection ratio check test circuit for table XVIII.

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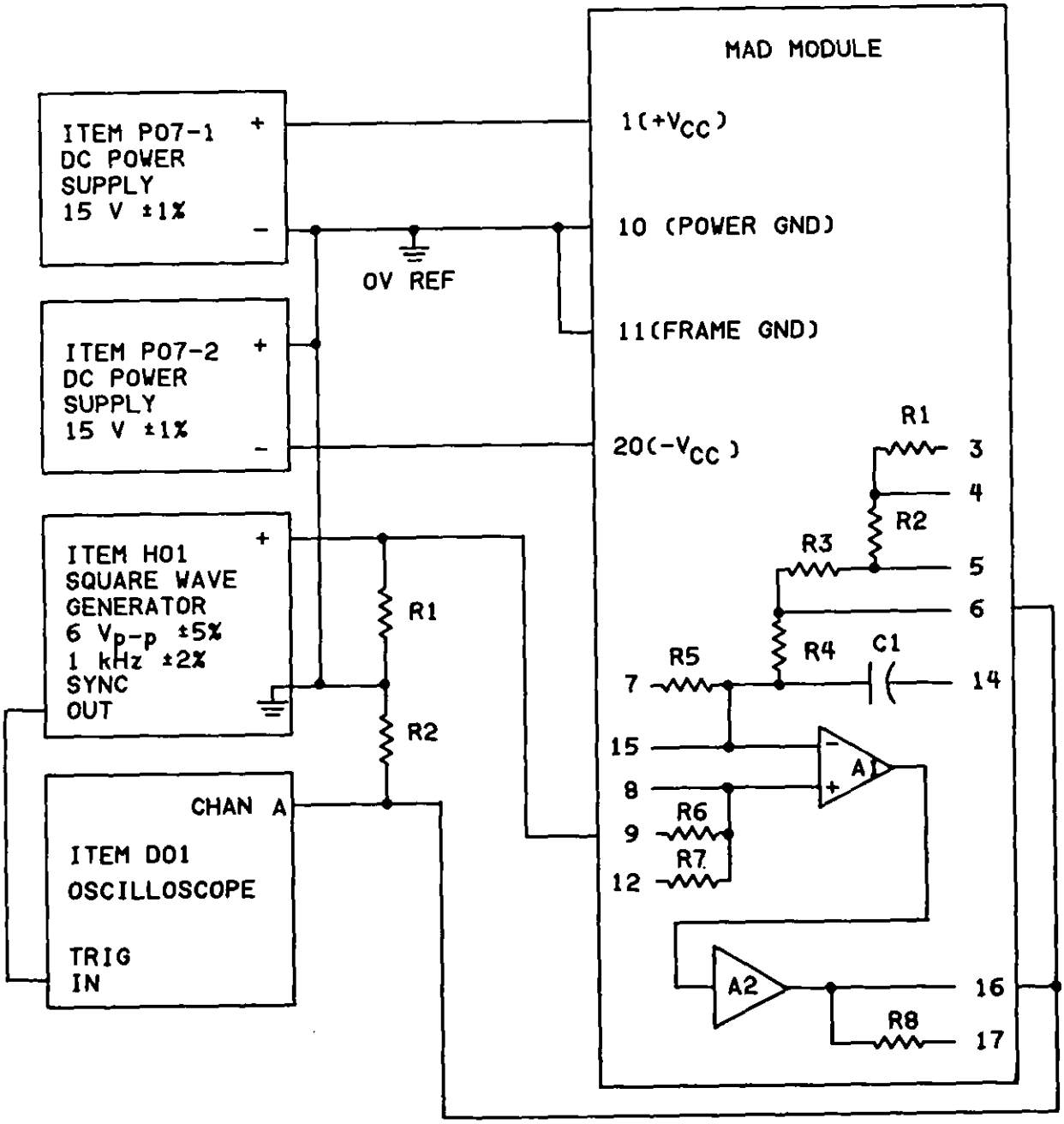
TABLE XIX. Slew rate check test. 1/

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
X1	Slew rate check (negative to positive)	1.0	13.3	µs	21, 22
X2	Slew rate check (positive to negative)	1.0	13.3	µs	21, 22

1/ This test verifies a slew rate of 300 mV/µs in accordance with the following relationship:

$$\text{Slew rate} = \frac{\Delta E_{out}}{\Delta t} = \frac{4}{\Delta t} \quad .$$

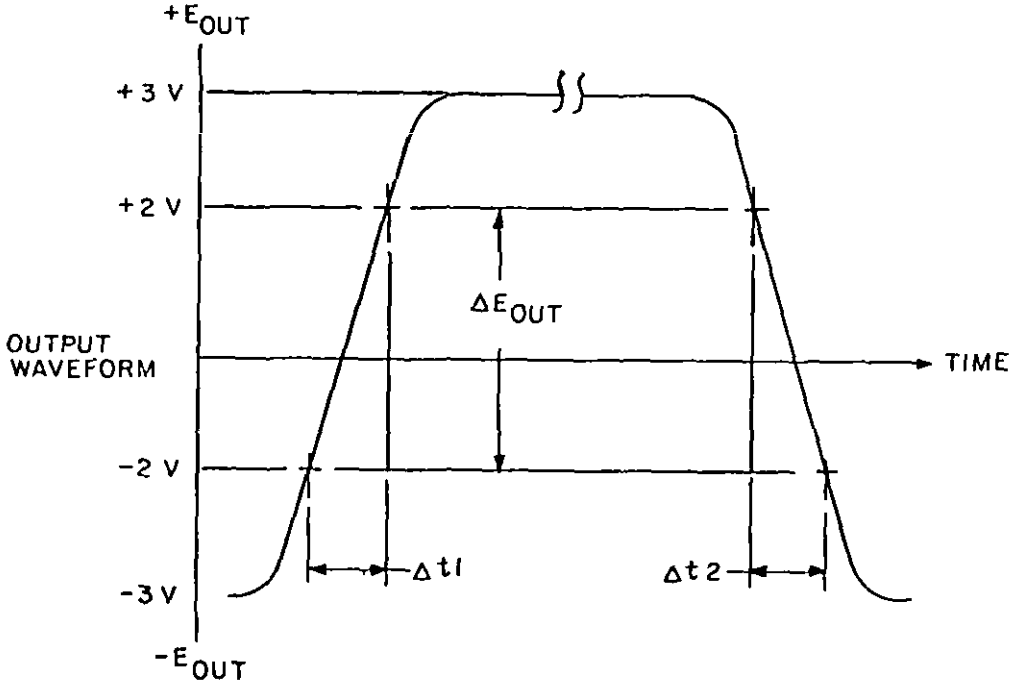
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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. R2 = 49.9Ω ±1%, 1 W.

FIGURE 21. Slew rate check test circuit for table XIX.

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NOTES
1. For test condition X1, measure $\Delta t1$ from the -2 V level to the +2 V level. The time in µs shall meet the table XIX limits.
2. For test condition X2, measure $\Delta t2$ from the +2 V level to the -2 V level. The time in µs shall meet the table XIX limits.

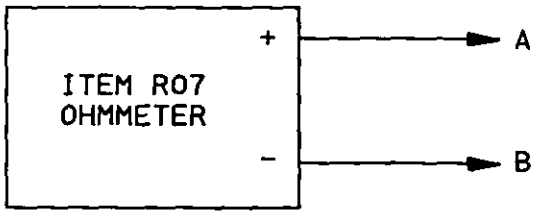
FIGURE 22. Slew rate check test circuit waveform for table XIX.

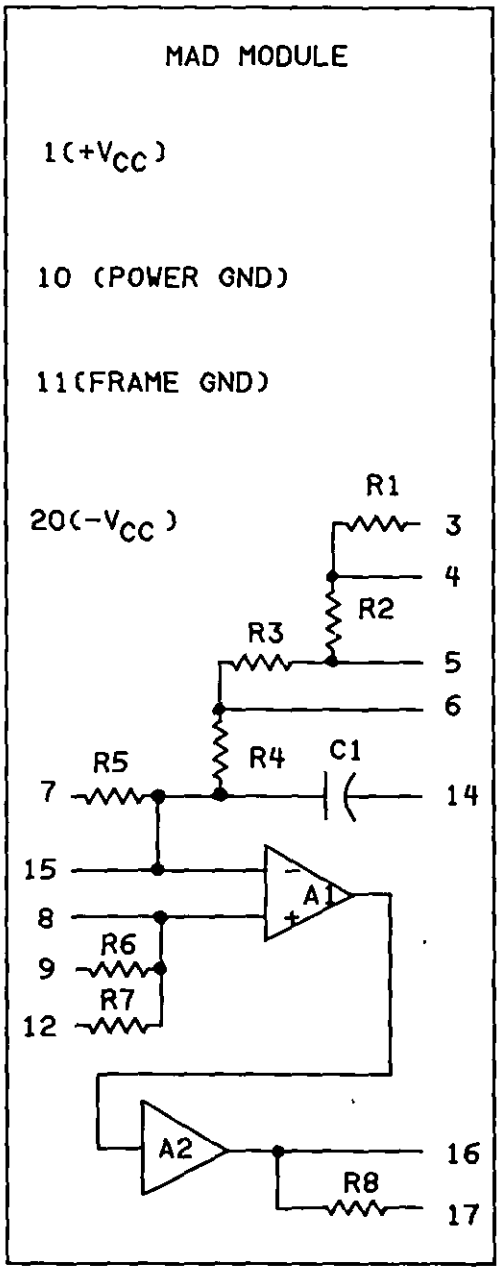
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TABLE XX. Frame continuity test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Y	Frame continuity	11	Frame	0	1	Ω	23

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NOTE: When measuring to the frame, penetrate the anodized finish of the module frame with a sharply pointed probe connected to the negative terminal.

* FIGURE 23. Frame continuity test circuit for table XX.

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